

MODULE 04 · DOMAIN 1 - DATA INGESTION AND TRANSFORMATION

Migration and connectivity

AWS DMS, DataSync, Transfer Family, AppFlow, Data Exchange, and how data reaches AWS at all.

DOMAIN 1 of 4 EXAM WEIGHT 34% MODULE 4 of 27

These questions are answered by the
shape of the source, not by the destination.

Source shape to service

THE SOURCE IS	REACH FOR
A relational database, ongoing	AWS DMS with change data capture
A relational database, one-off	A Glue job over a JDBC connection
Files on NFS, SMB or HDFS	AWS DataSync
Partners uploading to you	AWS Transfer Family
A SaaS application	Amazon AppFlow
Another company's dataset	AWS Data Exchange
An API you call	Lambda, Glue, or Step Functions

ON THE EXAM

Direction matters. DataSync pulls from a source you control. Transfer Family receives what others push.

When AppFlow has no connector

- ◆ **AWS Glue Marketplace connectors** cover many sources AppFlow does not.
- ◆ **Custom Glue connectors** can be built against a JDBC or Spark interface.
- ◆ **Glue connections** then make that source usable from any Glue job.
- ◆ Preferable to bespoke code, because the job still gets Glue's retries and logging.
- ◆ Still more effort than a managed AppFlow connector. Check AppFlow first.

ON THE EXAM

The order of preference is AppFlow, then a Glue connector, then custom code. Each step adds maintenance.

The phases of a database migration

01

Assess

What is there,
and how hard is
it to move?

02

Convert

Schema and
code, if the
engine is
changing.

03

Full load

Copy everything
that exists now.

04

CDC

Stream ongoing
changes while
the source stays
live.

05

Cut over

Switch
applications
once the target
has caught up.

ON THE EXAM

The reason CDC exists is the fourth box. The source keeps serving traffic while the copy catches up.

DMS is not only for migrations. It is a **continuous replication engine** that happens to be named for one use.

The three pieces of a DMS setup

- ◆ **Replication instance**, or DMS Serverless. The compute doing the work.
- ◆ **Source endpoint**. Where to read from, and the credentials to do it.
- ◆ **Target endpoint**. Where to write to.
- ◆ **Replication task**. Binds a source, a target and a migration type together.
- ◆ **Table mappings and transformations** select and rename objects in flight.

ON THE EXAM

One replication instance can host many tasks. That is how several databases consolidate into one target.

The three migration types

TYPE	USE IT WHEN
Full load	A one-off copy. The source can be frozen or is static
Full load plus CDC	The standard. Copy everything, then keep up with changes
CDC only	The target already has the data. You just need ongoing changes

ON THE EXAM

CDC requires the source database to be configured for it. That configuration is a frequent exam answer.

Replication instance or DMS Serverless?

REPLICATION INSTANCE

You size it

- Choose an instance class and storage
- Billed per hour while it exists
- Multi-AZ option for high availability
- Full control, including engine version

DMS SERVERLESS

AWS sizes it

- Capacity scales with the workload
- Billed per DMS capacity unit used
- No instance class to estimate
- Fewer knobs when you need to tune

THE TIEBREAKER

Unknown or variable migration volume favours Serverless. A long-running replication with known load can be cheaper provisioned.

DMS targets a data engineer cares about

- ◆ **Amazon S3.** Writes CSV or Parquet. The data lake ingestion path.
- ◆ **Amazon Redshift.** Direct into the warehouse.
- ◆ **Kinesis Data Streams and Amazon MSK.** Database changes as a stream.
- ◆ **Amazon OpenSearch Service.** Keeping a search index in sync.
- ◆ Plus the obvious relational targets, RDS, Aurora and databases on EC2.

ON THE EXAM

DMS to Kinesis turns a database into a change stream without touching the application.

DMS writing to Amazon S3

- ◆ **Output as CSV or Apache Parquet.** Parquet is the sensible choice for a lake.
- ◆ **Full load writes one set of files.** CDC appends change records after it.
- ◆ **Change records carry an operation column,** marking insert, update or delete.
- ◆ **The target is not a table.** It is a log of changes you must reconcile.
- ◆ **Apache Iceberg or Hudi** on the target side collapses that log into current state.

ON THE EXAM

DMS to S3 gives you a change log, not a current-state table. Something downstream has to apply the changes.

Same engine or different engine?

HOMOGENEOUS

Oracle to Oracle

- Schema moves across unchanged
- DMS alone is enough
- Native tooling option for MySQL and PostgreSQL
- Low risk, mostly mechanical

HETEROGENEOUS

Oracle to PostgreSQL

- Schema and code must be converted first
- Two steps: convert, then migrate
- Stored procedures may need manual work
- DMS Schema Conversion does the assessment

THE TIEBREAKER

Different engines means convert first, then move the data. One service does not do both in one step.

Three ways into the warehouse

APPROACH	CHOOSE IT WHEN
Zero-ETL integration	Supported pair, no transformation needed, lowest overhead
AWS DMS with CDC	Continuous sync, unsupported pair, or a non-Redshift target
Glue job over JDBC	Scheduled extract, and you need to transform on the way

ON THE EXAM

Ask two questions. Is the pair supported by zero-ETL, and does the data need reshaping on the way?

Where DMS tasks actually go wrong

- ◆ **CDC captures nothing.** Source logging not enabled. Fix the database.
- ◆ **Large objects truncated.** Limited LOB mode silently cuts at the size you set.
- ◆ **Task fails on a table with no primary key.** CDC needs a way to identify rows.
- ◆ **CDC latency climbing.** Target cannot keep up, or the instance is undersized.
- ◆ **Data validation** compares source and target row by row after migration.

ON THE EXAM

Limited LOB mode is fast and truncates. Full LOB mode is complete and slow. That trade-off is examinable.

Securing a DMS migration

- ◆ **Credentials in AWS Secrets Manager**, referenced by the endpoint.
- ◆ **Encryption at rest** with AWS KMS on the replication storage and the target.
- ◆ **SSL or TLS in transit** between DMS and both endpoints.
- ◆ **Data masking** can obscure column values before they reach the target.
- ◆ **Multi-AZ** gives a standby replication instance for long-running replication.

ON THE EXAM

Availability warning: AWS DMS Fleet Advisor reaches end of support on 20 May 2026.

Finishing a migration properly

- ◆ **DMS data validation** compares source and target rows and reports mismatches.
- ◆ **Let CDC latency reach near zero** before you cut over. That is the readiness signal.
- ◆ **Cut over** by stopping writes to the source, waiting for drain, then repointing applications.
- ◆ **Keep the task running** afterwards if you want a fallback path.
- ◆ **Do not delete the source** until validation has passed on the target.

ON THE EXAM

CDC latency approaching zero is what tells you the target has caught up. It is the metric cutover questions use.

A retailer runs a self-managed Oracle database on Amazon EC2 supporting its order system. The analytics team wants order data available in Amazon Redshift for reporting, refreshed within a few minutes, and the application team has refused any changes to the Oracle database or its application code.

A data engineer configures an AWS DMS task with the migration type set to full load plus CDC. The full load completes successfully and all existing rows appear in Redshift. No subsequent changes ever arrive, and the task remains in a running state without reporting an error.

What is the most likely cause?

- ☐ A The replication instance is undersized and cannot keep up with the change volume
- ☐ B Supplemental logging is not enabled on the source Oracle database
- ☐ C The target Redshift cluster requires a manifest file for incremental loads
- ☐ D The task should have used CDC only rather than full load plus CDC

◆ PAUSE AND DECIDE

What is the most likely cause?

- A

 Undersized replication instance
Would show rising CDC latency and changes arriving late. Nothing at all arriving is a different failure.
- B

 Supplemental logging not enabled
CDC reads the Oracle redo log. Without supplemental logging there is nothing for DMS to read, and no error to raise.
- C

 Redshift needs a manifest
Invented. DMS manages loading into Redshift itself, staging through S3 without your involvement.
- D

 Should have used CDC only
CDC only skips the full load. The full load worked. This would lose the existing rows and fix nothing.

THE SIGNAL

Full load succeeds, CDC silently does nothing, no error. That combination is always source-side logging.

For files, the question is **who initiates** the transfer, and whether the network can carry it.

AWS DataSync

- ◆ **An agent** runs on-premises to reach NFS, SMB, HDFS or object storage.
- ◆ **No agent needed** between AWS services, or from other clouds.
- ◆ **Handles the hard parts.** Verification, encryption, retries, incremental transfer.
- ◆ **Preserves metadata**, permissions and timestamps, which plain copying does not.
- ◆ **Targets** Amazon S3, Amazon EFS and Amazon FSx.

ON THE EXAM

DataSync transfers incrementally after the first run, so a scheduled task moves only what changed.

AWS Transfer Family

- ◆ **Managed SFTP, FTPS, FTP and AS2** endpoints in front of storage.
- ◆ **Backed by Amazon S3 or Amazon EFS.** Files land where your pipeline can see them.
- ◆ **External parties use their existing clients.** Nothing changes on their side.
- ◆ **Users managed by the service**, or authenticated against your own identity provider.
- ◆ **AS2** is for business-to-business document exchange, common in retail and logistics.

ON THE EXAM

The value is that partners keep using SFTP while the data lands directly in S3, with no server to run.

DataSync or Transfer Family?

DATASYNC

You pull

- You control the source system
- Scheduled or one-off transfer tasks
- Bulk migration of a file estate
- Metadata and permissions preserved

TRANSFER FAMILY

They push

- Third parties control the timing
- A standing endpoint, always available
- Protocol compatibility is the point
- Per-user access to prefixes

THE TIEBREAKER

If the question names an external party, it is Transfer Family. If it names your own file servers, it is DataSync.

The Snow family is being switched off

- ◆ **Snowmobile** retired in March 2024. The lorry is gone.
- ◆ **Snowcone** discontinued November 2024. Support ended November 2025.
- ◆ **Snowball Edge** closed to new customers on 7 November 2025.
- ◆ **All Snowball support ends 31 December 2026** in commercial Regions.
- ◆ AWS points you to **DataSync**, **AWS Data Transfer Terminal**, and **Outposts**.

ON THE EXAM

Study it anyway. Snowball remains a documented answer in exam-scope material while being unpurchasable.

One more option for file estates

- ◆ **AWS Storage Gateway, File Gateway.** An on-premises NFS or SMB mount backed by S3.
- ◆ Applications keep writing to a file share. The data lands in Amazon S3.
- ◆ Frequently accessed files are cached locally for low-latency reads.
- ◆ **Not a migration tool.** It is an ongoing hybrid access pattern.
- ◆ Contrast with DataSync, which copies and finishes.

ON THE EXAM

Applications that must keep using a file interface indefinitely means Storage Gateway.
A one-way move means DataSync.

A manufacturer receives daily production files from thirty external suppliers. Each supplier's system uploads to a company-managed SFTP server running on an Amazon EC2 instance, and a cron job then copies the files into Amazon S3, where a Glue job processes them.

The operations team patches and monitors the EC2 instance and has had two outages from full disks. Suppliers cannot change how they send files. The company wants to remove the server while keeping processing starting as soon as a file lands.

What should the company implement?

- ☐ A Replace the EC2 instance with AWS Transfer Family for SFTP writing directly to Amazon S3
- ☐ B Deploy AWS DataSync agents at each supplier site to pull the files into Amazon S3
- ☐ C Move the SFTP server to a larger EC2 instance in an Auto Scaling group with a bigger volume
- ☐ D Ask suppliers to upload directly to Amazon S3 using presigned URLs

◆ PAUSE AND DECIDE

What should the company implement?

- A

Transfer Family SFTP to S3
Same protocol for suppliers, no server to run, files land in S3 so an event notification starts Glue immediately. Also deletes the cron job.
- B

DataSync agents at supplier sites
Requires installing software in thirty companies you do not control. Wrong direction and organisationally impossible.
- C

Bigger instance, Auto Scaling
Keeps every operational burden they asked to remove, and SFTP servers do not scale horizontally cleanly anyway.
- D

Presigned URLs to S3
Would work technically and requires all thirty suppliers to change their systems. Explicitly ruled out.

THE SIGNAL

"Suppliers cannot change how they send files" is a protocol requirement. It eliminates B and D immediately.

Amazon AppFlow

- ◆ **Managed connectors** to Salesforce, ServiceNow, Zendesk, SAP, Slack and others.
- ◆ **Bidirectional.** Pull data out, or push data back into the SaaS application.
- ◆ **Filter, mask and validate** records in the flow, without code.
- ◆ **Run on demand**, on a schedule, or triggered by an event in the source.
- ◆ Only works with supported connectors. A niche SaaS tool may not be listed.

ON THE EXAM

AppFlow exists so nobody has to write and maintain OAuth token handling and pagination for a SaaS API.

AWS Data Exchange

- ◆ **A marketplace** for third-party datasets, free and paid.
- ◆ **Delivered into your account** as S3 objects, through APIs, or into Redshift.
- ◆ **Providers publish revisions**, and subscribers get them automatically.
- ◆ Removes the licensing, transfer and update mechanics of buying data.
- ◆ **Added to the in-scope service list** in exam guide version 1.1.

ON THE EXAM

Newly examinable. A question about acquiring an external dataset now has a named AWS answer.

When you have to call an API yourself

- ◆ **Lambda on a schedule** for small, periodic pulls. Mind the 15 minute limit.
- ◆ **Step Functions** when pagination, retries and rate limits need orchestrating.
- ◆ **Glue Python shell job** for a heavier pull that outlives Lambda.
- ◆ **Credentials in Secrets Manager**, never in the function's environment variables.
- ◆ **Rate limits** mean exponential backoff, not tighter retry loops.

ON THE EXAM

The exam guide names consuming data APIs as a skill. It usually appears as a least-effort comparison against AppFlow.

A media company stores customer records in Salesforce. The analytics team wants those records in Amazon S3 daily, with email addresses masked before they land, and with records for one region excluded entirely.

A data engineer has proposed an AWS Lambda function on an EventBridge schedule that authenticates to the Salesforce API, pages through results, applies the filtering and masking, and writes to S3. The team has limited capacity for ongoing maintenance.

What should the data engineer do instead?

- ☐ A Build the Lambda function as proposed, storing the Salesforce credentials in AWS Secrets Manager
- ☐ B Use Amazon AppFlow with a scheduled flow, applying the filter and mask in the flow configuration
- ☐ C Use AWS DMS with Salesforce as a source endpoint and Amazon S3 as the target
- ☐ D Subscribe to the Salesforce dataset through AWS Data Exchange

◆ PAUSE AND DECIDE

What should the data engineer do instead?

- A

 Build the Lambda as proposed
It works. It is also custom code owning auth, pagination and rate limits, for a team with no maintenance capacity.
- C

 DMS with Salesforce as a source
DMS migrates databases. Salesforce is a SaaS application, not a supported relational source.
- B

 AppFlow scheduled flow with filter and mask
A managed Salesforce connector, with filtering and masking as configuration. Every requirement, no code.
- D

 Data Exchange subscription
Data Exchange is for buying third-party datasets. Your own Salesforce data is not on a marketplace.

THE SIGNAL

Filtering and masking are AppFlow features. The question listed them as requirements to point you there.

Getting the network right

REQUIREMENT	THE MECHANISM
Private link to AWS over the internet	Site-to-Site VPN. Encrypted, uses your existing connection
Consistent bandwidth and latency	AWS Direct Connect. A dedicated physical circuit
Reach an AWS service privately from a VPC	Interface VPC endpoint, using AWS PrivateLink
Reach S3 or DynamoDB privately from a VPC	Gateway endpoint, added to the route table
A fixed public IP for a partner to allowlist	NAT gateway, or an Elastic IP on the endpoint

ON THE EXAM

Direct Connect gives predictability, not just speed. That distinction decides questions.

Allowlists and fixed addresses

- ◆ **Partners often allowlist source IPs** before they will accept your connections.
- ◆ **Lambda in a VPC gets an unpredictable address** unless you route it deliberately.
- ◆ **A NAT gateway with an Elastic IP** gives every outbound call one stable address.
- ◆ **Transfer Family endpoints** can be given static addresses for inbound allowlisting.
- ◆ Security groups control your side. Their allowlist controls theirs.

ON THE EXAM

An intermittently failing connection to a partner API, with no code changes, is often an unstable source address.

A logistics company must migrate 400 terabytes of historical archive files from an on-premises SMB file share into Amazon S3. The connection to AWS is a 1 Gbps link that is already carrying production traffic during business hours.

The files must retain their original permissions and timestamps. After the migration, a nightly transfer of roughly 200 gigabytes of new files must continue indefinitely.

Which approach best meets the requirements?

- ☐ A Use AWS DataSync for both the bulk migration and the ongoing nightly transfer, scheduling it outside business hours
- ☐ B Copy the files with the AWS CLI using multipart upload and a parallel sync script
- ☐ C Deploy AWS Storage Gateway File Gateway and let the cache populate over time
- ☐ D Use AWS Transfer Family with an SFTP endpoint and script the uploads

◆ PAUSE AND DECIDE

Which approach best meets the requirements?

- A

DataSync for both phases

Preserves SMB metadata, throttles to a bandwidth limit, transfers incrementally after the first run, and handles both requirements with one service.
- B

CLI sync script

Moves bytes and loses permissions and timestamps. Also code to write, schedule and monitor forever.
- C

File Gateway

Provides ongoing hybrid file access. It is not a bulk migration tool and does not move the archive.
- D

Transfer Family and scripts

Transfer Family receives what others push. Using it to send your own data is backwards, and metadata is still lost.

THE SIGNAL

Compare with scenario 2. There the external party made it Transfer Family. Here the source is yours, so it is DataSync.

The whole module in one table

SERVICE	ONE SENTENCE
AWS DMS	Relational databases, one-off or continuously, with CDC
AWS DataSync	You pull files from a source you control, metadata preserved
AWS Transfer Family	Others push files to you over SFTP, FTPS, FTP or AS2
AWS Storage Gateway	Ongoing hybrid file access, not a migration
Amazon AppFlow	Managed connectors to SaaS applications
AWS Data Exchange	Subscribe to datasets somebody else publishes
AWS Snowball	Offline bulk transfer. Being retired by the end of 2026

ON THE EXAM

If you can recite this table, most of the module's questions become fifteen-second answers.

Reading a migration question quickly

- ◆ **What is the source?** Database, files, SaaS, third party, or an API.
- ◆ **Who initiates?** You pull with DataSync. They push to Transfer Family.
- ◆ **One-off or ongoing?** Ongoing database sync means DMS with CDC.
- ◆ **Is the engine changing?** If so, convert the schema before moving data.
- ◆ **What is explicitly refused?** No source changes, no partner changes, no code.

ON THE EXAM

Next: choosing a transformation engine, including where large language models now fit.